Simulating Ground States and Dynamics of Small Quantum Many-Body Systems Using DMRG

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Outline

- **DMRG Algorithm Principles**
- **TEBD vs DMRG**
- **DMRG Applications**

How the density matrix renormalization group variationally optimizes a matrix product state.

Comparing two core MPS algorithms:
imaginary- time evolution vs. variational sweeping.

H_2 molecule and 1D hydrogen chain.
3.1 Single H_2 Molecule Ground State
Calculating bond length, energy, and electron correlation.
3.2 1-D Hydrogen Chain (H_{10}): Energy & Geometry
Finding the optimal dimerized configuration and proving Peierls dimerization.

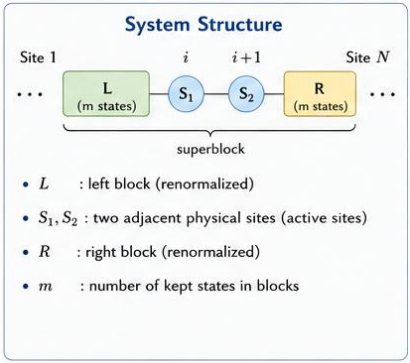
DMRG Algorithm Principles

1.1 Background and motivation

- Calculating **the ground states** of **quantum many-body systems** is a fundamental problem in condensed matter physics and quantum chemistry. For one-dimensional strongly correlated systems, **exact diagonalization (ED)** can provide benchmark results, but its computational complexity grows exponentially with system size, severely limiting the size of systems that can be studied.
- Since its introduction by **Steven R. White** in 1992, **the Density Matrix Renormalization Group (DMRG)** has become one of the most important numerical tools for solving one-dimensional strongly correlated systems. White's groundbreaking work shifted the approach to quantum many-body problems from traditional "diagonalizing large matrices" to "**constructing and optimizing the reduced density matrix between the system and the environment**."

1.2 Density Matrix Renormalization Group

- **Decomposition**（块分解）: Decompose the many-body wavefunction on a 1D lattice into a small "active" system and large "renormalized" environments.
- **Truncation**（截断）: Retain only the most significant states of the environments by diagonalizing the reduced density matrix.
- **Variational Optimization**（变分优化）: Iteratively improve the wavefunction by sweeping the active part across the lattice back and forth until convergence.



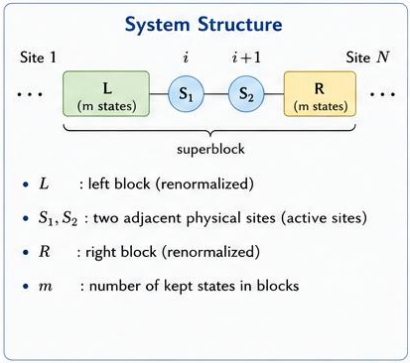
1.2 Density Matrix Renormalization Group

Goal: Reduce the dimension of blocks *L*and *R*while preserving accuracy.

$$\mathcal{H} = \mathcal{H}_{\mathcal{L}} \otimes \mathcal{H}_{\mathcal{S}_1} \otimes \mathcal{H}_{\mathcal{S}_2} \otimes \mathcal{H}_{\mathcal{R}}$$

$$|\Psi\rangle = \sum_{\alpha,i,j,\beta} \Psi_{\alpha i j \beta} \; |\alpha\rangle_L |i\rangle_{S_1} |j\rangle_{S_2} |\beta\rangle_R$$

where $\Psi_{\alpha i j \beta}$are the **probability amplitudes**



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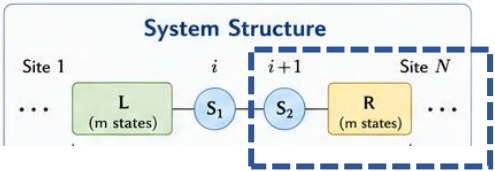


Truncation error

$$\epsilon = 1 - \sum_{i=1}^{m} \omega_i$$

Controlled by m.

Method

1. Construct the reduced density matrix for block L-S_1:

$$\boldsymbol{\rho_{LS_1}} = \mathbf{Tr}_{\boldsymbol{S_2R}} (|\Psi\rangle\langle\Psi|)$$

2. **Diagonalize $\boldsymbol{\rho_{LS_1}}$**:

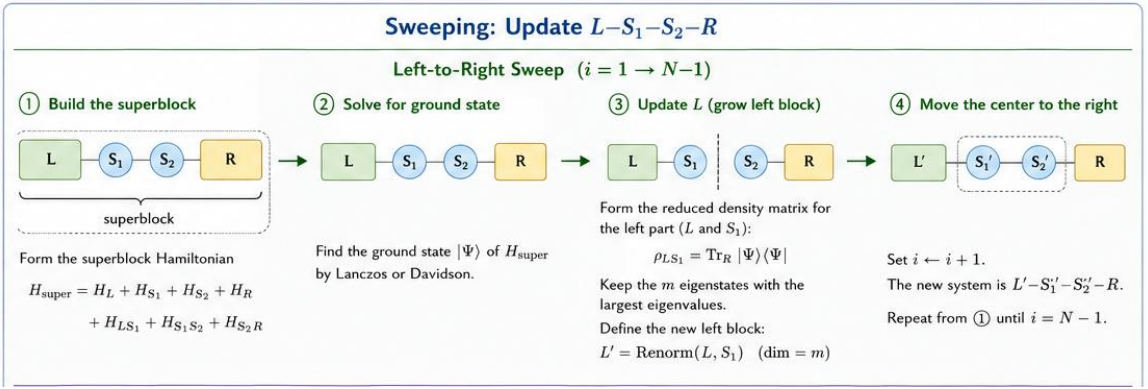
$$\rho_{LS_1}|\phi_k\rangle = w_k|\phi_k\rangle$$

where $w_k$are eigenvalues, $w_1 \geq w_2 \geq \cdots$

3. **Truncation**: Keep only the $\boldsymbol{m}$ most important states with largest w_k:

$$|\Psi\rangle \approx \sum_{k=1}^{m} \sqrt{w_k}\ |\phi_k\rangle_{LS_1} \otimes |\chi_k\rangle_{S_2R}$$

1.3 DMRG Algorithm



1.4 Algorithm Implementation

We study **the one-dimensional XXZ model** with open boundary conditions:

$$H = \sum_i \left(S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta S_i^z S_{i+1}^z \right)$$

- L: number of lattice sites
- S_i^α: spin-1/2 operators at site i
- Δ: anisotropy paramete

Goal: Find the **ground state energy** $\mathbf{E_0}$ and **wavefunction** $|\mathbf{\Psi_0}\rangle$

1.4 Algorithm Implementation

```
# Step 1 & 2: 构建超块并求解基态
H_super = self.build_superblock(left, right)
eigvals, eigvecs = eigh(H_super)
energy = eigvals[0].real
psi = eigvecs[:, 0]

# Step 3: 计算约化密度矩阵
dim_l, dim_r = left['dim'], right['dim']
psi_mat = psi.reshape(dim_l, dim_r)
rho = psi_mat @ psi_mat.conj().T
rho /= np.trace(rho)

# Step 4: 对角化并截断
rho_eigvals, rho_eigvecs = eigh(rho)
idx = np.argsort(rho_eigvals)[::-1]
chi = min(self.chi_max, len(rho_eigvals))
proj = rho_eigvecs[:, idx[:chi]]
```

```
# 初始化：两个单格点
left = self.init_site()
right = self.init_site()

# 逐步扩增到目标大小
for step in range(1, self.L // 2):
    left = self.enlarge(left)
    right = self.enlarge(right)

    left, energy, err_l = self.step(left, right)
    right, energy, err_r = self.step(right, left)

    current_size = 2 * (step + 1)
```

1.4 Algorithm Implementation

```
======================================================================
阶段 1: 无限 DMRG (系统逐步生长)
======================================================================

Step    系统大小          保留维度          能量                截断误差
----------------------------------------------------------------------
    能量: -1.61602540, 维度: 4, 截断误差: 3.33e-16
    能量: -1.61602540, 维度: 4, 截断误差: -2.22e-16
1       4             4             -1.61602540     3.33e-16
    能量: -2.49357713, 维度: 8, 截断误差: 3.33e-16
    能量: -2.49357713, 维度: 8, 截断误差: -2.22e-16
2       6             8             -2.49357713     3.33e-16
    能量: -3.37493260, 维度: 16, 截断误差: 1.33e-15
    能量: -3.37493260, 维度: 16, 截断误差: 1.11e-15
3       8             16            -3.37493260     1.33e-15

✓ 达到目标系统大小 8
```

```
基态能量:
  总能量 E_total = -3.3749325987
  每格点能量 E/L = -0.4218665748

参考值（热力学极限）: -0.4431
有限尺寸效应: 0.0212

计算时间: 0.08 秒
======================================
```

1.5 Modern DMRG Improvement: SVD

- Modern DMRG (especially within the **MPS** framework) directly performs **SVD** on the wavefunction matrix. **The two approaches are mathematically equivalent.**

Let Ψ be the ground state wavefunction reshaped into a matrix:

Traditional approach Construct left-block density matrix $\rho_L = \Psi\Psi^\dagger$

then diagonalize:

$$\rho_L = U\Lambda U^\dagger$$

Modern approach (SVD)

Directly perform SVD on the wavefunction:

$$\Psi = USV^\dagger$$

Substitute the SVD expression into the RDM formula: $\boldsymbol{\rho_L = (USV^\dagger)(VSU^\dagger) = US^2U^\dagger}$

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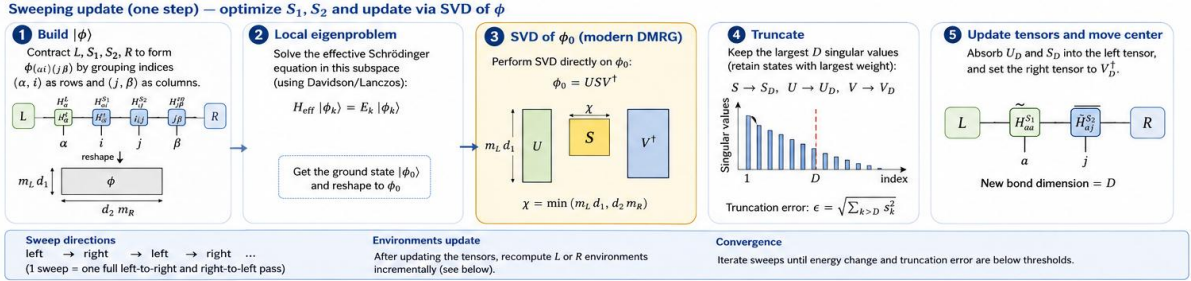
- Directly perform SVD on the wavefunction：$\Psi = USV^{\dagger}$



TEBD vs DMRG

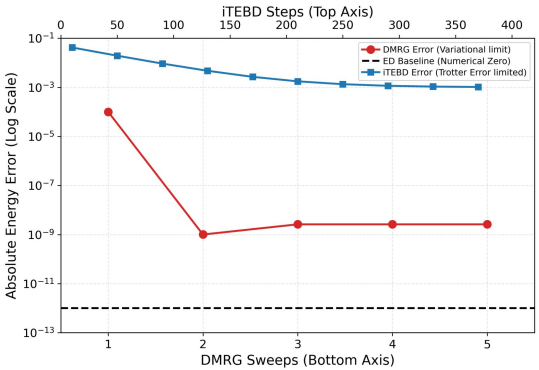
2.1 Time-Evolving Block Decimation

- **TEBD** is a tensor network algorithm for simulating **real / imaginary-time** evolution of 1D quantum systems. It represents the quantum state as a Matrix Product State (**MPS**) and updates local tensors using singular value decomposition (**SVD**). It is ideal for finite-size systems and short-time dynamics.
- **iTEBD** extends TEBD to infinite-size systems by assuming a uniform translational invariant MPS. It works directly in the thermodynamic limit using a unit cell representation. iTEBD is efficient for finding ground states of infinite systems and for studying **quantum quenches** in the thermodynamic limit.

2.2 Comparative Study of DMRG and iTEBD

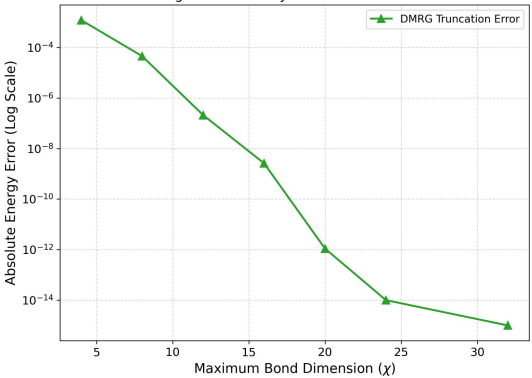
Ground State Energy Convergence（收敛性）

Figure 1: Ground State Energy Convergence Error



Accuracy vs. Bond Dimension

Figure 2: Accuracy vs. Bond Dimension



2.3 DMRG Core Advantages : Static Ground State Solving

1 No Trotter Error

- iTEBD relies on the Suzuki-Trotter decomposition of the time evolution operator, which inevitably introduces a **systematic error** proportional to the time step dt. DMRG completely bypasses time evolution, reformulating the problem as a **spatial variational optimization**. For a given bond dimension χ, DMRG searches for the **absolute minimum** of the energy within the variational manifold.

```
=== Method Comparison: Ground State Energy ===
ED Energy: -4.2580352073
iTEBD Energy: -4.2569375212 | Relative Error: 2.5779e-04
DMRG Energy: -4.2580352046 | Relative Error: 6.2643e-10
```

2 Fast Convergence & High Precision

- DMRG typically reaches machine-precision accuracy within just a few sweeps.

Computation of H-Chain

Goals

- Use DMRG to find H2’s ground state energy, to verify its accuracy.
- Find the optimal bond length for 1D equal-spacing H-chain and their ground state energy. Then Compare with independent H atoms to show the thermodynamic stability.
- Further analysis about the stability of 1D H-Chain.

Computational Methodology

- Main modules we use:
  - PySCF - RHF molecular orbitals
  - block2 - matrix product states (MPS) and DMRG
- Workflow:
  - Build molecule using coordinates and STO-3G/STO-6G basis set
  - Get 1e/2e integrals (h1e, g2e) & core energy
  - Initialize DMRG driver, MPO and random MPS
  - Sweep with increasing bond dimension
  - Converge energy & extract optimal geometry

Workflow example - H_2

```
mol = gto.M(atom="H 0 0 0; H 0 0 0.74", basis="sto3g", symmetry="d2h", verbose=0)
mf = scf.RHF(mol).run(conv_tol=1e-12)
```

- Molecule building:
  2 hydrogen atoms at (0,0,0) and (0,0,0.74)
  using STO-3G basis set (a minimal basis set, fast but not very accurate)

- RHF(Restricted Hartree-Fock):
  compute the initial molecule orbit, which is a simplified model that DMRG will start from

Workflow example - H_2 - Cont.

- Setup Hamiltonian:

  Hamiltonian of H_2 is:

$$\boldsymbol{H} = -\frac{1}{2}\nabla_1^2 - \frac{1}{2}\nabla_2^2 - \frac{1}{|\mathbf{r}_1 - \mathbf{R}_a|} - \frac{1}{|\mathbf{r}_1 - \mathbf{R}_b|} - \frac{1}{|\mathbf{r}_2 - \mathbf{R}_a|} - \frac{1}{|\mathbf{r}_2 - \mathbf{R}_b|} + \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} + \frac{1}{|\mathbf{R}_a - \mathbf{R}_b|}$$

- DMRG cannot directly process the differential equations in continuous space, need to project H onto the molecular orbitals calculated by RHF

  One-electron integrals h1e:

$$\boldsymbol{h_{pq}} = \int \psi_p^*(\mathbf{r}) \left[-\frac{1}{2}\nabla^2 - \frac{1}{|\mathbf{r} - \mathbf{R}_a|} - \frac{1}{|\mathbf{r} - \mathbf{R}_b|} \right] \psi_q(\mathbf{r})\, d\mathbf{r}$$

  and Two-electron integral g2e:

$$\boldsymbol{g_{pqrs}} = \iint \psi_p^*(\mathbf{r}_1)\psi_q(\mathbf{r}_1) \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} \psi_r^*(\mathbf{r}_2)\psi_s(\mathbf{r}_2)\, d\mathbf{r}_1 d\mathbf{r}_2$$

Workflow example - H_2 - Cont.

- Finally, we use these operators to write the Hamiltonian in a discrete form, and this process is called the **second quantization** of the Hamiltonian.

$$\hat{H} = \sum_{pq} h_{pq}\, \hat{a}_p^\dagger \hat{a}_q + \frac{1}{2} \sum_{pqrs} g_{pqrs}\, \hat{a}_p^\dagger \hat{a}_r^\dagger \hat{a}_s \hat{a}_q + E_{core}$$

- The process mentioned above is done through:

```
ncas, n_elec, spin, ecore, h1e, g2e, orb_sym = itg.get_rhf_integrals(mf, ncore=0, ncas=None, g2e_symm=8)
```

- The variables are:

  Number of orbitals (ncas): 2
  Number of electrons (n_elec): 2
  Total spin (spin): 0
  Nuclear repulsion energy, core energy (ecore): 0.7151043391 Hartree

Workflow example - H_2 - Cont.

- Initialize DMRG: Specify symmetry and other parameters to boost calculation.

```
driver = DMRGDriver(scratch="./tmp", symm_type=SymmetryTypes.SU2, n_threads=4)
driver.initialize_system(n_sites=ncas, n_elec=n_elec, spin=spin, orb_sym=orb_sym)
mpo = driver.get_qc_mpo(h1e=h1e, g2e=g2e, ecore=ecore, iprint=0)
ket = driver.get_random_mps(tag="GS", bond_dim=20, nroots=1)
```

- Set DMRG sweep configurations:
  - bond_dims : 2 is enough for H2, 100 is an absolutely safe value.
  - noises : add noise to jump out of local minimum.
  - thrds : cut off thresholds for DMRG sweeps.

```
n_sweeps = 8
bond_dims = [100] * n_sweeps
noises    = [1e-4, 1e-5, 1e-6, 1e-7, 0, 0, 0, 0]
thrds     = [1e-10] * n_sweeps
```

Workflow example - H_2 - Cont.

- Run DMRG and output energy:

```
energy_DMRG = driver.dmrg(mpo, ket, n_sweeps=n_sweeps,
                          bond_dims=bond_dims, noises=noises, thrds=thrds, iprint=0)
print(f"H2 ground state energy = {energy_DMRG:.12f} Hartree")
```

- The result:

  H2 ground state energy = -1.137283834489 Hartree

Benchmark on H_2 Molecule

- Bond length = 0.74Å, using STO-3G basis set
  ground state energy = -1.137283834489 Hartree
- Compare with exact energy from PySCF.FCI :

```
fci_e = fci.direct_spin1.kernel(h1e, g2e, ncas, n_elec)[0] + ecore
print(f"PySCF FCI参考值:    {fci_e:.12f} Hartree")
print(f"与DMRG计算结果的误差:    {energy_DMRG - fci_e:.2e} Hartree")
```

- FCI reference value : -1.137283834489 Hartree
- The error is : -8.88e-16 Hartree
- The result of DMRG approach is consistent with FCI result.

- But is it close enough to real world?

Benchmark on H_2 Molecule - Cont.

- The ground state energy of H2 is -1.1745 Hartree
- Our result is - 1.137 Hartree, still has a notable error.

- The error partly comes from the STO-3G basis - simple but inaccurate
- Change the basis to cc-pVTZ - a basis set including polarization functions

```
mol = gto.M(atom="H 0 0 0; H 0 0 0.74", basis="cc-pVTZ", symmetry=True, verbose=0)
```

- We get the result as -1.156572416521 Hartree, which is lower and also closer to the CBS(Complete Basis Set) limit.

- Besides, the DMRG intermediate results show that the accurate result is achieved just after the 1st sweep, the H2 case is still too simple for it.

Analysis on Equal-Spacing H_n Chains

- We need to compute energy of H-chain and compare it with independent H atoms to show the thermodynamic stability.
- We set up equal-spacing H-chain with N atoms (N=2,4,6,8,10,12,14).
- For each H-chain, scan the bond length in range and run DMRG to find the minimal ground state energy.

- The molecule definition setup is done by:

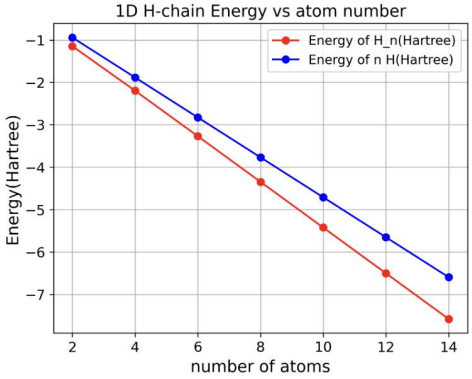
```python
def build_coords(n_atoms, bond_length):
    coords = []
    x = 0.0
    for i in range(n_atoms):
        coords.append(['H', (x, 0, 0)])
        x += bond_length
    return coords
```

```python
coords = build_coords(n_atoms=n_atoms,bond_length=bond_length_angstrom)


mol = gto.M(atom=coords, basis='sto6g', symmetry='c1', verbose=0)
mf = scf.RHF(mol).run(conv_tol=1e-12)
```

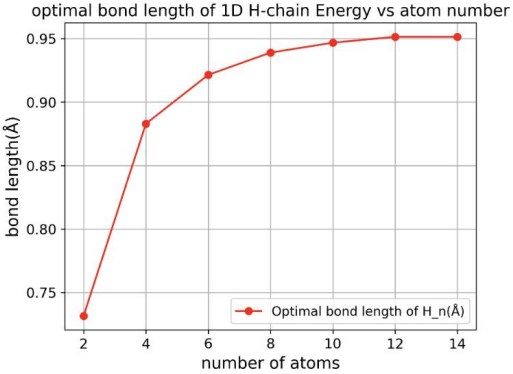
Analysis on Equal-Spacing H_n Chains - Cont.

- For all H-chains, we plot the energy v.s. atom number and optimal bond length v.s. stom number as following(with bond length eps = 0.02Å):



- For all H-chains, the energy is lower than the sum of the same number of independent H atoms, which means they are stable in thermaldynamics.

Analysis on Equal-Spacing H_n Chains - Cont.



- The optimal bond length grows with the growing number of atoms, showing that the strength of bond is weakening as the chain grows. Especially for N=2 and N=4.

- But is equal-spacing ideal enough for 1D H-chains?
  - The drastic changes in energy around N=2,4 suggest that the H-chains might lower its total energy by forming units with smaller spacing.

Peierls Dimerization

- Existing studies on **Peierls dimerization** show that for one-dimensional metals, the system spontaneously shifts from a higher-energy 'metallic state' (with **uniform spacing**) to a lower-energy 'insulating state' (a dimerized structure with **alternating long and short bonds**).
- Whether dimerization occurs depends on the interplay of two types of energy:
  - electronic energy: The band gap opens, and the occupied state energy decreases.
  - elastic energy: Deviation from the equilibrium bond length, cost of lattice distortion.
- Total energy: $$\Delta E(\delta) = \underbrace{A\delta^2}_{Elastic\ penalty} - \underbrace{B\delta^2 \ln\frac{1}{\delta}}_{Electronic\ gain}$$
- For small δ, we have large $\ln\frac{1}{\delta}$ and $\Delta E < 0$, so any one-dimensional half-filled metal will inevitably dimerize at zero temperature.

Considering Dimerized H_{10} Configuration

- In a 1D H-chain, each hydrogen atom contributes one electron, exactly filling half of the energy band, forming a 'half-full' **metallic state**. So lower energy is feasible through dimerization.
- Building dimerized molecule:

```python
def build_dimerized_coords(n_atoms, R_short, R_long):
    coords = []
    x = 0.0
    for i in range(n_atoms):
        coords.append(['H', (x, 0, 0)])
        if i % 2 == 0:
            x += R_short
        else:
            x += R_long
    return coords
```

Coarse Initial Scan（初始粗扫描）

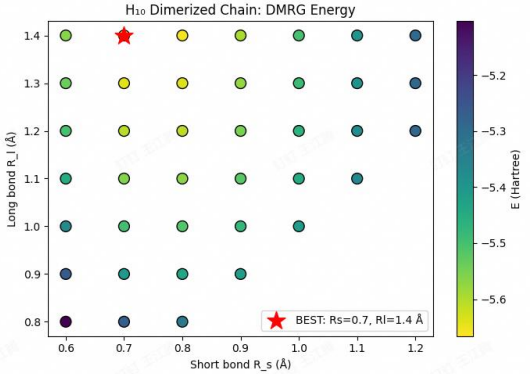
```python
n_atoms = 10
R_short_range = np.arange(0.6, 1.3, 0.1)  # 短键：0.6 ~ 1.2 Å
R_long_range  = np.arange(0.8, 1.5, 0.1)  # 长键：0.8 ~ 1.4 Å
```

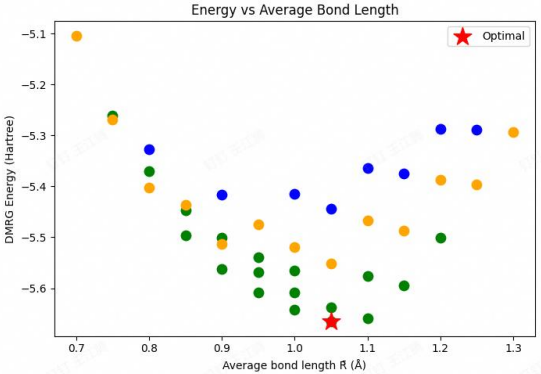
```python
results = []
for R_s in R_short_range:
    for R_l in R_long_range:
        if R_l <= R_s:
            continue  # 跳过无意义的组合（长键必须大于短键）
        try:
            energy, e_rhf = compute_dimerized_energy(
                R_s, R_l, n_atoms=10, basis='sto6g', verbose=0
            )
            results.append({
                'R_short': R_s, 'R_long': R_l,
                'E_DMRG': energy, 'E_RHF': e_rhf,
                'E_corr': energy - e_rhf
            })
            print(f"Rs={R_s:.1f} Å, Rl={R_l:.1f} Å → E_DMRG = {energy:.10f} Hartree")
        except Exception as e:
            print(f"Rs={R_s:.1f}, Rl={R_l:.1f} 计算出错: {e}")
```

Conclution：

Considering the Dimerized Configuration Calculation for H_{10}

H_{10} Peierls 二聚化扫描结果（粗精度）
最优短键长： R_s* = 0.70 Å
最优长键长： R_l* = 1.40 Å
平均键长： R̄ = 1.05 Å
最低能量： E_dim = -5.6656960951 Hartree
二聚化参数： δ ≡ (R_l - R_s)/R̄ = 0.667

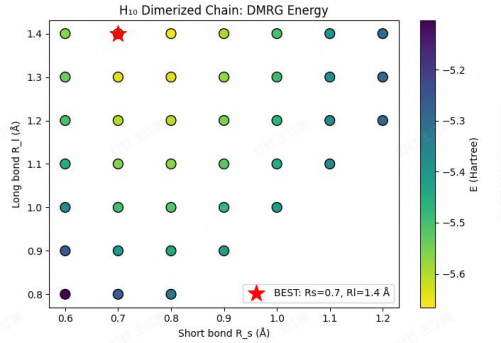


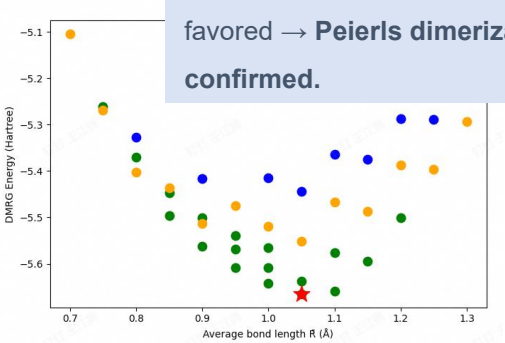


```
H₁₀ Peierls 二聚化扫描结果（粗精度）
最优短键长:   R_s* = 0.70 Å
最优长键长:   R_l* = 1.40 Å
平均键长:     R̄   = 1.05 Å
最低能量:     E_dim = -5.6656960951 Hartree
二聚化参数:   δ ≡ (R_l - R_s)/R̄ = 0.667
```

- Minimum energy ≈ -5.67 Hartree – lower than the uniform chain (all bonds equal).
- Diagonal positions (equal bond lengths) show higher energy.

So uniform distribution is not energetically favored → **Peierls dimerization effect is confirmed.**





Conclusion for H_{10}

- If we compare R_s with the bond length of H_2, which is about 0.74Å, we'll find that the structure of dimerized H_{10}-chain looks like 5 hydrogen molecules.
- Energy of 5 H_2 molecules: -5.6864191724 Hartree
- Energy of H_{10}-chain: -5.6879675956 Hartree
- Error: -0.0015484231 Hartree (0.0272%)

- We can conclude that H_{10}-chain is in fact 5 H_2 molecules.

Main Conclusions

- The bond length of a 1D equally spaced H-chain increases as the number of atoms goes up, and the bonding gets weaker.
- Although a 1D equally spaced H-chain can be thermodynamically stable, in reality it's just metastable and tends to dimerize.
- Calculations for H_{10} show that after dimerization, the 1D H-chain is basically just independent H_2 molecules, and long one-dimensional hydrogen chains can't really exist stably.

Simulating Ground States and Dynamics of Small Quantum Many-Body Systems Using DMRG

Thank you for your attention.